Contents

Where Data Comes From, and What It Costs

> **Supplementary reading — Lesson 2** Estimated reading time: 25 minutes Not examinable. It is, however, the constraint that will decide which of your ideas you are allowed to build.

Why a computer scientist should read this

Lesson 2 treated a dataset as something that arrives: a CSV file with columns that need cleaning. That is how it looks from inside a notebook, and it is the one view that will never be enough professionally.

Three things this adds.

Somebody produced every row, and it cost something. Usually money, sometimes labour under conditions worth knowing about, occasionally the privacy of a person who never agreed to be in your training set.

In the European Union, what you may do with data is law, not preference. You will work under the General Data Protection Regulation (GDPR). It decides whether a project is legal before it decides whether it is good, and reading it after the model is built is the expensive order.

The dataset's history is part of the result. A model trained on data gathered under one set of assumptions carries those assumptions into every prediction it makes.

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1. The regulation you will actually work under

Regulation (EU) 2016/679 — the GDPR — has applied since 25 May 2018 across the Union. It governs the processing of **personal data**: anything relating to an identified or identifiable living person.

That definition is much broader than most engineers expect. A name obviously qualifies. So does an IP address, a device identifier, a customer number, or any combination of ordinary columns that together single somebody out.

1.1 Six principles, in the terms a developer meets them

Article 5 lists the principles. Four of them change how you write code.

Purpose limitation. Data collected for one purpose may not simply be reused for another. The churn dataset gathered to send invoices is not automatically available for training a marketing model. This is the principle most often broken by accident, because a data warehouse makes everything look equally available.

Data minimisation. Collect what the purpose requires, and no more. It is the opposite instinct to the one machine learning encourages — "keep every column, the model might find something" is a defensible engineering heuristic and an indefensible legal position.

Accuracy. Inaccurate personal data must be corrected or erased. A model trained on stale records is not merely worse; it may be unlawful.

Storage limitation. Keep it no longer than the purpose needs. "Forever, in case it is useful" is not a retention policy.

The remaining two — lawfulness and fairness, and integrity and confidentiality — are the ones lawyers spend most time on and engineers least.

1.2 You need a lawful basis before you start

Article 6 gives six, and you must pick one *before* processing: consent, performance of a contract, legal obligation, vital interests, public task, or legitimate interests.

Two practical notes. **Consent is the weakest basis**, not the strongest — it must be freely given, specific, informed and revocable, and if it is withdrawn your basis disappears. And **legitimate interests** requires a documented balancing test against the rights of the person, not a shrug.

1.3 The categories you must not touch casually

Article 9 singles out **special category data**: racial or ethnic origin, political opinions, religious beliefs, trade union membership, genetic and biometric data, health, sex life and sexual orientation. Processing is prohibited by default, with narrow exceptions.

The trap for machine learning is that you can process this data **without collecting it**. A postcode plus a shopping history is a serviceable proxy for ethnicity in many European cities. The regulation is concerned with effects, and a model that infers a special category has processed one.

1.4 Anonymous, or merely pseudonymised?

This distinction decides whether the regulation applies at all, and it is routinely got wrong.

Pseudonymised data — identifiers replaced by keys, with the mapping kept somewhere — is *still personal data*, and the GDPR still applies in full. Hashing an email address is pseudonymisation, not anonymisation: the hash is stable, so it still singles a person out.

Anonymous data, which can no longer be attributed to a person by any means reasonably likely to be used, falls outside the regulation entirely.

Genuine anonymisation is much harder than it looks, and the literature on re-identification is a long series of demonstrations that it failed. Small combinations of ordinary attributes turn out to be surprisingly unique in a population, and a dataset released as anonymous can often be re-identified by joining it against another one.

The engineering lesson: treat “we anonymised it” as a claim requiring evidence, especially when the evidence is your own.

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2. The part of the pipeline nobody puts on a slide

Supervised learning needs labels, and labels are made by people.

ImageNet — the dataset whose 2012 result opened the modern era — contains roughly fourteen million labelled images. They were labelled through Amazon Mechanical Turk by tens of thousands of workers across many countries, over about two years. The compute and the architecture get the credit; the labelling was the larger part of the effort and cost.

That pattern has not changed, it has grown. Content moderation, medical image annotation and the human feedback behind current language models are all performed by people, frequently in lower-income countries, frequently on piecework rates, and frequently on material that is unpleasant or worse to look at. Reporting on the conditions of moderation and annotation work has been consistent enough over the past decade that a practitioner should not be surprised by it.

Two consequences that are about engineering rather than ethics alone.

Label quality is a function of labour conditions. Piecework paid by the item rewards speed, and speed produces a particular kind of noise: the easy cases are right and the ambiguous ones are guessed. That is not random noise, and it does not average out.

Annotator disagreement is data, not a nuisance. Where two competent people disagree about a label, the task itself is ambiguous — and a model trained to a single “gold” answer will be confidently wrong on exactly those cases. Datasets that publish per-annotator labels are more useful than those that publish only the majority.

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3. Data cascades: why upstream problems compound

Practitioner studies of applied machine learning describe a recurring pattern often called **data cascades**: problems introduced at collection are invisible at the time, survive every downstream step, and surface only in deployment, where they are most expensive to fix.

The shape is always the same. Nobody was responsible for the data itself. The collection was done by whoever had access, the cleaning by whoever inherited it, and the modelling by somebody who assumed both had been done properly.

Two habits are worth more than any technique in lesson 2:

- **Write down where every column came from, and when.** A column whose provenance nobody can state is a column nobody can defend.
- **Talk to whoever generated the data.** Fifteen minutes with the person who operates the sensor or fills in the form routinely explains a distribution that a week of exploratory analysis only describes.

4. What this means for your project

Concretely, for the final project of this course and for the first year of your working life:

- **Prefer datasets with a documented licence and provenance.** “Found on Kaggle” is not provenance; the Kaggle page usually links to the actual origin.
- **Ask what the collection purpose was**, and whether your use matches it.
- **Assume the data is personal until you have established otherwise**, rather than the reverse.
- **Record what you did.** Not for the marker — for the version of you in eighteen months who has to explain the model to somebody who did not build it.

Where to look next

- **Regulation (EU) 2016/679** itself. It is more readable than its reputation; Articles 5, 6, 9 and 22 are the ones to know, and the recitals explain the reasoning.
- Your national data protection authority publishes practical guidance — in Italy, the **Garante per la protezione dei dati personali**.
- **Datasheets for Datasets** (Gebru et al.) proposes a standard set of questions every dataset should answer about its own origin. Worth reading once and then using as a checklist.
- **Data and its (dis)contents** (Paullada et al.) surveys what dataset practices in machine learning research actually look like.